MAGNETOSTATICS

The Lorentz Force Law

When charges move, they produce a magnetic as well as an electric field.

$\vec{F}_{mag} = Q(\vec{v} \times \vec{B})$ → Lorentz force law.

This gives us: $\vec{F}_{net} = Q[\vec{E} + (\vec{v} \times \vec{B})]$

It is important to note that magnetic forces do no work:

$dW_{mag} = F_{mag} \cdot d\vec{l} = Q(\vec{v} \times \vec{B}) \cdot \vec{v}\,dt = 0$

* Problems:

3.) Beams of electrons, charge q, mass m:

a.) Pass the beam through $\perp$ $\vec{E}$ and $\vec{B}$, adjusted electric field for 0 deflection. What was particle v?

$v = \frac{E}{B}$ ✓

b.) $\vec{E} = 0$, find R_{beam} due to field. Find q/m in terms of E, B and R.

We know: $\frac{mv^2}{R} = qvB \rightarrow \frac{q}{m} = \frac{v}{RB}$

Assuming particles are entering this region with: $v = E_{old}/B$

$\frac{q}{m} = \frac{E_{old}}{RB^2}$ ✓

For a current carrying wire:

$F_{mag} = \int I(d\vec{l} \times \vec{B})$

For a current flowing on a surface: $\vec{K} = \frac{d\vec{I}}{dl_\perp} = \sigma\vec{v}$ (surface current density.) $\longrightarrow F_{mag} = \int(\vec{K} \times \vec{B})\,da$

Similarly, for volume: $\vec{J} = \frac{d\vec{I}}{da_\perp} = \rho\vec{v}$ (volume current density.) $\longrightarrow \vec{F}_{mag} = \int(\vec{v} \times \vec{B})\rho\,d\tau = \int(\vec{J} \times \vec{B})\,d\tau$

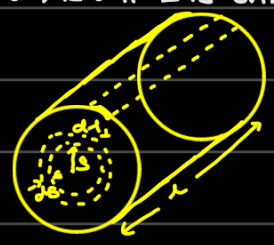
Now, total current crossing a surface S: $I = \int_S \vec{J} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{J})\,d\tau = -\frac{d}{dt}\int_V \rho\,d\tau$

This gives rise to the continuity eq^n: $\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$ → charge conservation.

* Problems:

5.) Current I flows down a wire of radius a:

a.) K? if I is uniformly distributed over the surface:



we know: $\vec{K} = \frac{dI}{dl_\perp} \longrightarrow \vec{K} = \frac{I}{2\pi a}$ ✓

How much I is present in an area element: $l\,dl_\perp$

Total current: I, Total area: $2\pi a l$

$\therefore$ I in $l\,dl_\perp = \frac{I}{2\pi a l} \cdot l\,dl_\perp = \frac{I}{2\pi a}\,dl_\perp$

b.) $J(s) \propto 1/s$, what is $J(s)$?

$I = \int J(s).da = \int_0^a \frac{k}{s} 2\pi s\,ds = k.2\pi a \quad . \therefore k = \frac{I}{2\pi a} \longrightarrow J(s) = \frac{I}{2\pi a} \cdot \frac{1}{s}$ ✓

7.) For a charge and current configuration in a volume V, show that: ${}_v\int \vec{J}\,d\tau = d\vec{p}/dt$

we know that $\vec{p} = \int \vec{r}'\rho(\vec{r}')\,d\tau'$

Time derivative: $\frac{d\vec{p}}{dt} = \frac{d}{dt}\int \vec{r}\rho\,d\tau = -\int \frac{\partial \rho}{\partial t}\vec{r}\,d\tau$

Now: ${}_v\int \nabla.(x\vec{J})\,d\tau = {}_s\int x\vec{J}.da = 0$

And: ${}_v\int \nabla.(x\vec{J})\,d\tau = \int_v J_x\,d\tau + \int_v x(\nabla.\vec{J})\,d\tau = \int_v J_x\,d\tau + \int \frac{\partial \rho}{\partial t}x\,d\tau = 0$

Thus: ${}_v\int J_x\,d\tau = \frac{dP_x}{dt}$ Hence Proved.

The Biot-Savart Law

Magnetostatics imposes the condition: $\nabla.\vec{J} = 0$

Magnetic Field of a steady current is given by Biot-Savart Law:

Permeability of free space.

$\vec{B}(r) = \frac{\mu_0}{4\pi}\int \frac{\vec{I} \times \hat{r}}{r^2}\,dl' = \frac{\mu_0 I}{4\pi}\int \frac{dl' \times \hat{r}}{r^2}$

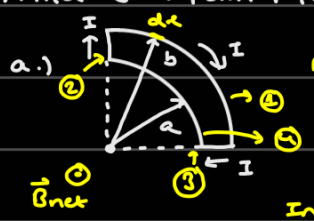
For surface and volume currents:

$\vec{B}(r) = \frac{\mu_0}{4\pi}\int \frac{\vec{K}(r') \times \hat{r}}{r^2}\,da' \; ; \; \vec{B}(r) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(r') \times \hat{r}'}{r^2}\,d\tau'$

* Problems:

9.) Find $\vec{B}$ at point P for each of the configurations shown below:

a.)



Biot-Savart Law: $\frac{\mu_0 I}{4\pi}\int \frac{dl \times \hat{r}}{r^2}$

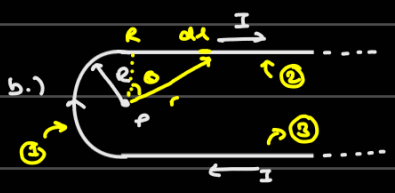
In wires ②, ④, there is no field generation, since $dl \times r = 0$.

In wire ①: $dl \times r = b\,dl \sin\alpha = b\,dl$ $(\because \alpha = 90°)$

$$\therefore \vec{B}_1 = \frac{\mu_0 I}{4\pi}\int_0^{\frac{\pi b}{2}} \frac{dl}{b^2} = \frac{\mu_0 I}{4\pi}\,\frac{\pi b}{2b^2} = \frac{\mu_0 I}{8b}$$

If wire ④ carries current I: $\vec{B}_2 = \frac{\mu_0 I}{8a}$; $\vec{B}_{net} = \frac{\mu_0 I}{8}\left(\frac{1}{a} - \frac{1}{b}\right)$

b.)



$\vec{B}_1 = \frac{\mu_0 I}{4R}$ (in) ✓

Now fields due to wires ② and ③ will be equal, so all we have to find is $\vec{B}_2$:

$$2\times\vec{B}_2 = 2\times\frac{\mu_0 I}{4\pi}\int \frac{dl \times \hat{r}}{r^2} = \frac{\mu_0 I}{2\pi} \text{ (in)}$$ ✓

Divergence and Curl of B

The line integral of B is always given by: $\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$

Applying stokes theorem: $\int (\nabla\times\vec{B})\cdot d\vec{a} = \mu_0 \int \vec{J}\cdot d\vec{a} \longrightarrow \nabla\times B = \mu_0\vec{J}$

This method of deriving

Magnetic Vector Potential

Just as $\nabla \times \vec{E} = 0$ allowed us to establish a scalar potential, $\nabla \cdot \vec{B} = 0$ allows for a vector potential:

$\vec{B} = \nabla \times \vec{A} \longrightarrow \nabla \cdot \vec{A} = 0$ This is chosen by choice (Gauge symmetry)

Basically, the vector potential is fixed by the curl but there is nothing telling us about the divergence, so we allow for no divergence.

With this condition: $\nabla^2 \vec{A} = \mu_0 \vec{J} \longrightarrow \vec{A}(r) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{r} d\tau'$ Assuming J goes to 0 at infinity.

Typically, vector potential follows the current.

There is no simple theoretical interpretation of the vector potential and it is not as useful as the scalar potential but it still finds theoretical importance.

The magnetic field (similar to the electric field) suffers a discontinuity at a surface. However, with the magnetic field the difference is tangential, not perpendicular.

$\vec{B}_{above} - \vec{B}_{below} = \mu_0(\vec{K} \times \hat{n})$ → can be proved using ampere's law.

In vector potential speak:

$$\frac{\partial \vec{A}_{above}}{\partial n} - \frac{\partial \vec{A}_{below}}{\partial n} = -\mu_0 \vec{K}$$

Also, magnetic dipole moment is given by: $\vec{m} \equiv I \int d\vec{a} = I\vec{a}$